Table A5 Pipeline pressure loss (vacuum)

British Standard Size Tube BS EN 1057: R250, Table X		Distance from source (m) at 59 kPa (450 mm Hg) for 1.3, 2.6, 3.9, 6.5 kPa (10, 20, 30, 50 mm Hg) pressure loss																
Outside Diameter (mm)	Pressure loss (kPa)	8	15	30	61	91	122	152	183	213	244	274	305	335	366	396	427	457
		Free air flow rate (L/min)																
12	1.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	2.6	47	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	3.9	60	40	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	6.5	82	55	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
15	1.3	59	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	2.6	89	59	40	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	3.9	113	76	51	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	6.5	153	103	69	46	–	–	–	–	–	–	–	–	–	–	–	–	–
22	1.3	173	116	78	52	41	–	–	–	–	–	–	–	–	–	–	–	–
	2.6	260	174	117	79	62	53	46	42	–	–	–	–	–	–	–	–	–
	3.9	330	222	149	100	79	67	59	53	49	45	42	40	–	–	–	–	–
	6.5	445	301	203	137	108	92	81	73	67	62	57	54	51	49	46	45	43
28	1.3	350	236	159	106	84	71	63	56	51	48	44	42	40	–	–	–	–
	2.6	525	353	238	160	127	107	94	85	78	72	67	63	60	57	54	52	50
	3.9	666	448	303	204	161	137	120	108	99	92	86	81	76	73	69	66	64
	6.5	900	607	412	278	220	187	164	148	135	125	117	110	104	99	95	91	87
35	1.3	637	427	288	193	153	130	114	102	94	87	81	76	72	69	65	63	60
	2.6	947	638	431	290	230	195	171	154	141	131	122	115	109	103	99	95	91
	3.9	1198	808	548	369	293	248	218	197	180	167	156	147	139	132	126	121	116
	6.5	1614	1091	743	503	399	339	298	269	246	228	213	200	190	180	172	165	158
42	1.3	1074	724	488	328	260	220	194	174	160	148	138	130	123	117	111	107	103
	2.6	1598	1079	731	493	391	331	291	262	240	222	208	196	185	176	168	161	155
	3.9	2016	1363	926	626	497	422	371	334	306	283	265	249	236	224	214	205	197
	6.5	2706	1833	1254	851	677	574	506	456	417	387	361	340	322	306	293	280	270
54	1.3	2191	1480	1001	674	535	453	399	359	329	304	284	268	253	241	230	220	212
	2.6	3246	2196	1493	1010	802	681	599	540	494	458	428	403	381	363	346	332	319
	3.9	4083	2766	1889	1281	1019	865	762	687	629	582	545	513	485	462	441	423	406
	6.5	5448	3699	2549	1737	1384	1176	1037	935	856	794	742	699	662	630	601	576	554
76	1.3	5521	3773	2563	1733	1377	1169	1029	927	849	786	735	692	655	623	595	570	548
	2.6	8070	5563	3807	2586	2058	1749	1541	1389	1273	1179	1103	1038	983	936	894	857	823
	3.9	10041	6968	4801	3274	2609	2219	1957	1765	1617	1499	1402	1320	1250	1190	1137	1090	1048
	6.5	13166	9233	6439	4421	3533	3009	2655	2396	2197	2037	1906	1796	1701	1619	1547	1483	1426
108	1.3	12874	9140	6543	4552	3732	3280	2879	2628	2433	2276	1919	2036	1941	1858	1785	1712	1641
	2.6	18207	12874	9235	6578	5274	4552	4071	3716	3441	3219	3035	2879	2745	2628	2525	2422	2325
	3.9	22494	15905	11374	8114	6509	5657	5030	4592	4251	3976	3750	3557	3391	3247	3119	2992	2870
	6.5	29238	20675	14708	10520	8445	7343	6538	5968	5526	5169	4873	4623	4408	4220	4055	3889	3730

Examples:

1 122 m of 28 mm pipe would carry 71 L/min of free air per minute with a pressure loss of 10 mm Hg (1.3 kPa), or 187 L/min with a loss of 50 mm Hg (6.5 kPa).
ie: $122/122 \times (71/71)^2 \times 1.3$

2 A flow of 120 L/min in 122 m of 28 mm pipe would result in pressure loss of 2.99 kPa. ie: $122/122 \times (120/137)^2 \times 3.9$

3 140 m of 28 mm pipe would carry 90 L/min with a pressure loss of 2.20 kPa. ie: $140/152 \times (90/94)^2 \times 2.6$